

quantisation and quantisation-aware training are two approaches. While post-training quantisation applies after model training, quantisation-aware training incorporates quantisation during the training process.

Edge-specific model architectures. Designing architectures tailored for edge devices can enhance efficiency. Lightweight models such as MobileBERT and TinyBERT are optimised for resource-constrained environments. These models maintain high performance while reducing computational overhead, making them suitable for edge deployment.

Hardware acceleration. Edge devices can leverage specialised hardware accelerators such as GPUs (graphics processing units), TPUs (tensor processing units), and NPUs (neural processing units). These accelerators are designed to handle AI workloads efficiently, providing substantial performance improvements over general-purpose CPUs.

Use cases of language models on edge devices

Voice assistants. Edge-based voice assistants enable real-time speech recognition and natural language understanding, providing users with instant responses and actions. This is particularly beneficial in scenarios where low latency is critical, such as smart homes and automotive applications. Language models on edge devices can enhance the functionality of smart home systems by providing more sophisticated and intuitive control through voice commands. From adjusting lighting and temperature to managing security systems and household appliances, these models make home automation more accessible and user-friendly.

Industrial automation. In industrial settings, edge-based language models can facilitate better human-machine interaction. Workers can use voice commands to control machinery, access technical

documentation, and receive real-time assistance with complex tasks. This can lead to increased efficiency, safety, and productivity in manufacturing and other industrial operations.

Also, integrating language models into IoT devices allows for intelligent automation. For example, smart thermostats can understand user preferences and adjust settings accordingly. Even systems such as washing machines can initiate predictive maintenance based on natural language inputs.

EDGE-BASED LANGUAGE MODELS HAVE REVOLUTIONISED HOW WE INTERACT WITH SMARTPHONES. BY LEVERAGING ADVANCED NATURAL LANGUAGE PROCESSING CAPABILITIES, THESE MODELS CAN UNDERSTAND AND RESPOND TO VOICE COMMANDS, ENHANCING THE USER EXPERIENCE.

Healthcare applications. Edge-based language models can revolutionise healthcare by enabling real-time, natural language interactions between patients and medical devices. These models can assist in patient monitoring, provide instant medical advice based on symptoms described, and facilitate seamless communication with healthcare providers. By operating on edge devices, they ensure data privacy and reduce latency, which is critical in medical emergencies. Language models on edge devices can assist healthcare professionals by providing instant access to medical information, patient records, and

diagnostic tools. This can improve decision-making and patient care, especially in remote or resource-limited settings.

Gaming and virtual reality. Edge-based language models enhance augmented reality and virtual reality experiences by enabling natural language interactions within virtual environments. This can be applied in gaming, training simulations, and virtual meetings, creating more immersive and interactive experiences. Players can communicate with game characters and navigate virtual environments using natural language, making the gaming experience more engaging and realistic.

Smart retail. Language models on edge devices can revolutionise the retail experience by providing personalised shopping experiences. Smart kiosks and interactive displays can understand and respond to customer queries in natural language, recommend products based on preferences, and even facilitate seamless checkouts. This enhances customer engagement and streamlines the shopping process.

Autonomous vehicles. In autonomous vehicles, edge-based language models can interpret and respond to voice commands, enhancing safety and convenience. Passengers can interact with the vehicle's AI system to control navigation, entertainment, and climate settings, creating a more intuitive and user-friendly experience.

Security and surveillance. Integrating language models into security systems can improve threat detection and response times. Edge devices equipped with these models can analyse audio inputs in real time to identify suspicious activities or unauthorised access attempts. This proactive approach to security ensures swift action and enhances overall safety.

Education and e-learning. Language models on edge devices